

HARSH VIKRAM SINGH

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EDUCATION

B.Tech in Computer Science and Engineering

Vellore Institute of Technology, Chennai | 7th Semester

Expected Graduation: 2027 | CGPA: 8.65 / 10

TECHNICAL SKILLS

Languages: C, C++, Python, Java, JavaScript, TypeScript, MySQL,

Web & Frameworks: HTML, CSS, React.js, Node.js, Express.js, FastAPI, REST API, GraphQL, Tailwind CSS, Bootstrap

AI & ML: OpenAI API, Google Gemini, XGBoost, Scikit-learn, ChromaDB, RAG, Prompt Engineering, LLM APIs

Blockchain: Smart Contracts, NFT Standards, On-chain Transaction Handling

Tools & Platforms: Git, GitHub, VS Code, Postman, Vercel, Render, Docker, Linux, Socket.IO, Docker, Kubernetes

PROJECTS

CodeFlow — Lead Developer

Live: codski.qzz.io | github.com/vikramhls/codeflow

- Built a collaborative, gamified codebase intelligence and bug bounty platform with AI-powered insights, 3D repository mapping, and an issue-bidding system.
- Developed full-stack architecture using TypeScript, React, Node.js with a RAG-based Repository Intelligence Assistant powered by ChromaDB vector embeddings.
- Deployed live at codski.qzz.io with a separate backend API on Render; handled system architecture, AI integrations (LLM + embeddings), and feature delivery end-to-end.

FinProtect AI – Financial Fraud Detection — Solo Project

github.com/vikramhls/financial-fraud-insight

- Built a real-time financial fraud detection platform combining XGBoost ML models with Google Gemini LLM to detect, explain, and classify fraudulent transactions.
- Developed a FastAPI + SQLAlchemy async backend with role-based JWT authentication for Customer and Risk Analyst roles.
- Designed an interactive React + TailwindCSS dashboard with an AI Chat Analyst enabling natural-language querying of transaction data.

Cyber Secure Vault — Lead Developer

Live: cyber-secure-vault-tdbu.vercel.app

- Built and deployed a secure file storage app with AES-256 encryption and role-based authentication using Node.js, Express.js, and React.js; deployed on Vercel.

NFT Rental Platform — Blockchain Developer

github.com/vikramhls/NFT_rental

- Developed a blockchain-based NFT rental system with smart contract logic for secure digital asset lending, rental mechanisms, and on-chain transaction handling.
- Built the full TypeScript stack covering contract interaction, rental lifecycle, and NFT ownership management.

Doctor Multilingual Assistance App — Group Project

github.com/vikramhls/doctor-lingual-helper

- Created a real-time video consultation tool with speech-to-text, translation, and text-to-speech for cross-language doctor-patient communication.
- Integrated Socket.IO, Vosk, and ElevenLabs APIs to enable seamless multilingual interactions from audio capture to translated voice output.

ACHIEVEMENTS & LEADERSHIP

- 2-Star Coder on CodeChef** (Username: basic_song_19) – Consistent competitive programming and problem-solving.
- LeetCode** – Solved 350+ algorithmic problems across core data structures and algorithms.
- Student Coordinator – Food Stalls, VIT Vibrance** – Coordinated food stall logistics and vendor operations for one of VIT's largest cultural events, ensuring efficient execution throughout the festival.
- Head of Finance, Sangam Cultural Club** – Managed budgets for 5+ large-scale events totaling over ₹1,00,000.
- Glitchathon Hackathon Participant** – Developed a full-stack campus portal in 24 hours under competitive conditions.

ADDITIONAL

- Proficient in Git, GitHub, and deploying real-time full-stack applications on cloud platforms (Vercel, Render).
- Experienced in building AI-integrated systems using LLM APIs, vector databases, and machine learning pipelines.
- Strong leadership, communication, and problem-solving skills demonstrated through team projects and club management.
- Passionate about building scalable, real-world solutions at the intersection of AI and full-stack development.